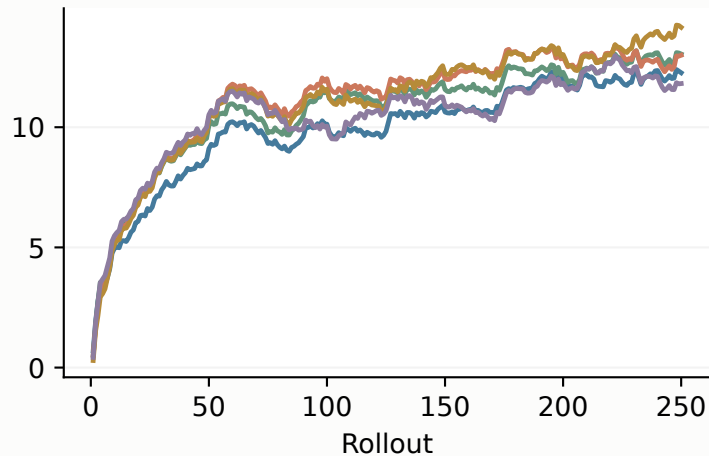
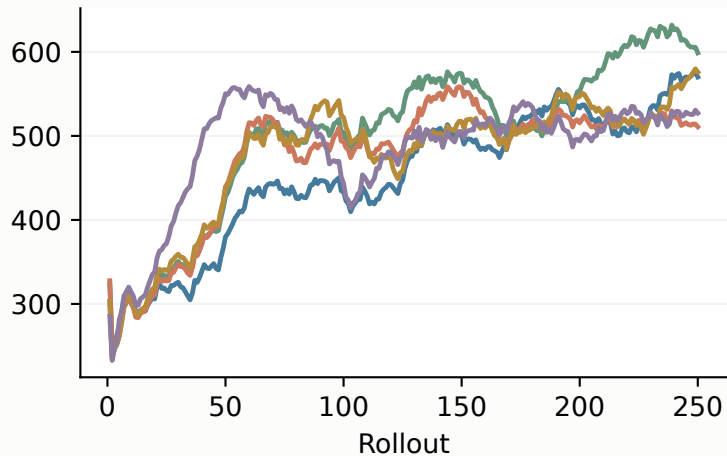


Qwen3-14B Base + SFT: λ and random-direction ablations

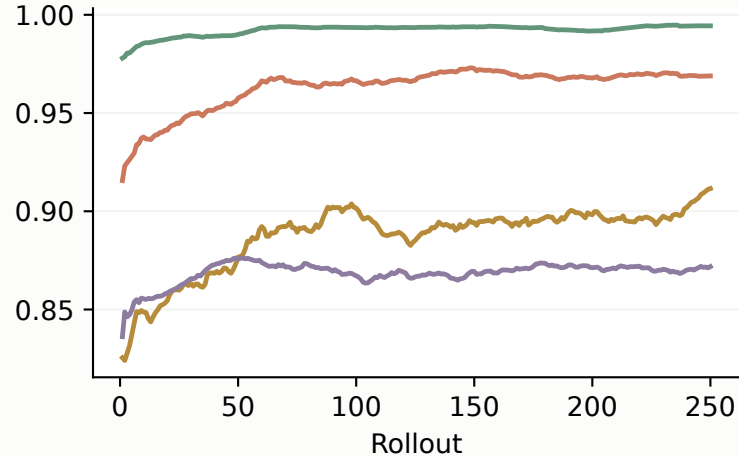
Length-adjusted reward



Response tokens



Credit ESS ratio



GRPO VPO $\lambda=2$ VPO $\lambda=4$ VPO $\lambda=8$ Random direction, $\lambda=4$